

Validity Analysis: GSM8K

Target context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Overall risk: **HIGH**

Deployment Context

We are a US edtech company building a tutoring module for American students. When a student gets an arithmetic word problem wrong, our system should identify where in their worked solution the error occurred and explain the mistake. The system receives the original word problem plus the student's step-by-step work (which may contain errors), and produces a diagnostic explanation. We want to evaluate the LLM's ability to analyze and diagnose errors in arithmetic reasoning.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	2	Significant gaps	high
Input Content ■	2	Significant gaps	high
Input Form ✓	3	Moderate gaps	high
Output Ontology ■	1	Serious concern	high
Output Content ■	1	Serious concern	high
Output Form ✓	2	Significant gaps	high
Average	1.8		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GSM8K is a problem-solving benchmark — it tests whether models can produce correct grade-school math word problem solutions, scored by binary final-answer correctness. The deployment is a diagnostic system that must analyze student-authored solutions, locate the earliest error and downstream errors, classify error types according to a five-plus-category US-curriculum taxonomy, and generate teacher-register explanations differentiated by grade level. The directional mismatch is severe on every HIGH-priority dimension: input ontology lacks an error taxonomy and grade-level metadata; input content is professionally authored clean prose with none of the student-register notation the deployment must parse; output ontology is binary final-answer correctness with no error-location, error-type, or

pedagogical-acceptability label space; output content is annotated by freelance contractors with no domain-educator involvement. The MR-GSM8K empirical result that top GSM8K solvers (>80% accuracy) collapse to MR-Scores <0.02 on error localization quantifies the gap as not merely structural but operationally severe. Partial alignments — shared text/English modality, USD currency, US-customary unit incidence, and the multi-step chain-of-thought solution format — are real but secondary. GSM8K is at best a final-answer sanity check; it cannot serve as the primary evaluation instrument for a diagnostic deployment.

Practical Guidance

What This Benchmark Measures

GSM8K measures a model's ability to *produce* a correct final numerical answer to a clean, professionally authored grade-school-level word problem when given the problem alone. It exercises multi-step chain-of-thought arithmetic reasoning (a strength for input_ontology) and verifies the final number against contractor-agreed ground truth. For the deployment, this provides only a baseline competency check: 'can this model do the underlying arithmetic if it had to?'

Construct Depth

Shallow with respect to the deployment's construct. The construct of interest is *diagnostic meta-reasoning over student work*, which has at least four sub-constructs: (1) parsing non-standard student notation, (2) localizing the earliest error, (3) classifying error type against a curriculum-aligned taxonomy, and (4) generating teacher-register explanation. GSM8K touches sub-construct (0) — does the model know the arithmetic at all — and none of (1)–(4). The MR-GSM8K finding of near-zero error-localization performance from GSM8K-strong models [WEB-7, WEB-25] is direct empirical evidence that depth on the deployment construct is essentially absent.

What Else You Need

Substantial supplementation required across all six dimensions. For IO/OO: ProcessBench [WEB-14] for earliest-error-step identification, MR-GSM8K [WEB-7] for error-reason generation as a sub-task, BEA 2025 Shared Task [WEB-16] for a four-dimensional tutor-response rubric, and a deployment-built error-type taxonomy aligned to the five categories plus US-curriculum subtypes. For IC: DrawEduMath [WEB-9, WEB-10] or Daheim et al. (2024) [WEB-13] for student-authored work distribution; ideally a deployment-collected corpus of grades 3–6 student work. For OC: an in-house teacher-annotated evaluation set with grade-differentiated pedagogical acceptability criteria (JEDM precedent [WEB-17]; DrawEduMath QA layer). For OF: rubric-based or LLM-judge evaluation along the BEA 2025 dimensions (Mistake Identification, Mistake Location, Providing Guidance, Actionability) plus human teacher review for register. GSM8K should be retained only as a baseline final-answer sanity check, not as a primary instrument.

ID	Evidence
WEB-7	MR-GSM8K: top LLMs >80% on GSM8K collapse to MR-Scores <0.02 on error localization (https://github.com/JIA-Lab-research/MR-GSM8K)
WEB-9	DrawEduMath teacher-annotated student work (CCSS-aligned) (https://arxiv.org/abs/2501.14877)
WEB-10	DrawEduMath benchmark site (https://drawedumath.org/)
WEB-13	Daheim et al. (2024) ~1K student-reasoning chains with first-error annotation (https://arxiv.org/pdf/2407.09136)
WEB-14	ProcessBench: 3,400 cases with human-annotated earliest erroneous step, including 400 GSM8K-level cases (https://arxiv.org/pdf/2412.06559)
WEB-16	BEA 2025 Shared Task: four-dimensional tutor-response evaluation framework (https://arxiv.org/pdf/2505.18549)
WEB-17	JEDM 8,307 problems / 193,187 student responses / 1,296 teachers with 5-point scale + free-text feedback — precedent for teacher-validated ground truth at scale (https://jedm.educationaldatamining.org/index.php/JEDM/article/download/858/258)
WEB-25	EmergentMind summary corroborating GSM8K→meta-reasoning collapse (https://www.emergentmind.com/topics/gsm8k)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment is HIGH-priority on input ontology because it requires diagnosing at least five error categories (misapplied operation, arithmetic slip, wrong setup, incomplete chain, plus US-curriculum-specific subtypes). GSM8K's task ontology is organized around a single task — solving grade-school math word problems — with no error-type subtaxonomy [Q1, Q11, Q22]. Its only category-like distinctions are methodological (finetuning vs. verification, solution-level vs. token-level verifier) [Q27, Q28, Q30, Q72, Q73], not diagnostic. Dataset inspection confirms every record has only {question, answer} with a final number after `####` and no error-type metadata (DATASET-D1, DATASET-D2). Additionally, a non-trivial portion of sampled problems is above the grades 3–6 CCSS-M band — algebraic variables, simple interest, ratio/proportion (DATASET-D8, DATASET-D9, DATASET-D10, DATASET-D11) — introducing construct-irrelevant variance for a grade 3–6 diagnostic evaluation. ASDiv offers grade-level/problem-type annotations [WEB-5] but does not provide student-error taxonomy. MR-GSM8K reframes GSM8K for error localization [WEB-7], but the base GSM8K ontology itself remains unsuited to the deployment.

ID	Evidence
Q1	"we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems." (p.1)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q22	"Similar to LogiQA, which requires a mix of reading comprehension and logical reasoning, GSM8K's main difficulty lies in both properly interpreting a question and reasoning through the steps to solve it." (p.4)
Q27	"We investigate two methods to solve problems in GSM8K: finetuning and verification." (p.5)
Q28	"Finetuning, our baseline method, uses the same language modeling objective as the generative pretraining in GPT-3 (Brown et al., 2020)." (p.5)
Q30	"In contrast, verification consists of sampling multiple high temperature solutions, assigning each solution a score, and outputting the highest ranked solution." (p.5)
Q72	"We can either train verifiers to make a single scalar prediction conditioned on the entire generated solution, or to make a scalar prediction after each token in the solution." (p.9)
Q73	"By default, we choose the latter, training verifiers to make predictions after each token." (p.9)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Clean, multi-step prose word problem requiring 4 sequential arithmetic operations
D2	On Tuesday Buddy has $30/2 = \ll 30/2=15 \gg 15$ baseball cards. On Wednesday Buddy has $15+12 = \ll 15+12=27 \gg 27$ baseball cards. — Well-formed natural language solution with calculator annotation format; no student notation
D8	Let x represent the number of Chevys; Fords: $3+2x$; Buicks: $4(3+2x)=12+8x$; Total: $x+3+2x+12+8x=301$; $11x+15=301$ — Uses algebraic variables and simultaneous-style equation — above elementary grade band (grades 3–6)
D9	If she earns an interest of 10% in the first year, her savings account increases by $10/100 * \$600 = \$\ll 10/100*600=60 \gg 60$. In the second year, she earns the same amount of interest, which $\$60 + \$60 = \$\ll 60+60=120 \gg 120$ — Simple interest calculation — above typical grades 3–6 curriculum; note solution applies simple (not compound) interest correctly
D10	Let N be the original price each friend was going to pay. $10N=6(N+8)$; $10N=6N+48$; $4N=48$; $N=\ll 12=12 \gg 12$ — Algebraic equation setup with variable N — outside grades 3–6 scope

D11	The total ratio of the coins they both have is $10+45 = \ll 10+45=55 \gg 55$. The fraction of the ratio representing the number of coins that Amalie has is $45/55$, and since the total number of coins they both have is 440, Amalie has $45/55 \cdot 440 = \ll 45/55 \cdot 440=360 \gg 360$ coins. — Ratio and proportion problem — grade 6 borderline, but not typical grades 3–5
WEB-5	ASDiv provides problem-type taxonomy and grade-level metadata but no student-error taxonomy (https://aclanthology.org/2020.acl-main.92.pdf)
WEB-7	MR-GSM8K: top LLMs >80% on GSM8K collapse to MR-Scores <0.02 on error localization (https://github.com/JIA-Lab-research/MR-GSM8K)

Strengths

- Problems consistently require 2–6 step chain-of-thought arithmetic reasoning, matching the kind of multi-step student work the deployment must analyze ([DATASET-D1](#), [DATASET-D16](#), [DATASET-D25](#)).
- The Socratic config decomposes solutions into explicit sub-question steps, providing a structural parallel to step-indexed analysis [[DATASET-D23](#)].
- GSM8K explicitly targets 'elementary concepts' and shares design principles with ASDiv [[Q11](#), [Q18](#)], so part of the ontology overlaps with grades 3–6 arithmetic.

ID	Evidence
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q18	"We share those design principles in the creation of GSM8K." (p.4)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Clean, multi-step prose word problem requiring 4 sequential arithmetic operations
D16	If the bakery can make 12 pies, this means there would be $12 * 3 = \ll 12 \cdot 3=36 \gg 36$ pie pieces. For all the pieces the bakery would make $36 * 4 = \$ \ll 36 \cdot 4=144 \gg 144$. The cost of making 12 pies is $12 * 0.5 = \$ \ll 12 \cdot 0.5=6 \gg 6$. That means the bakery would make $144 - 6 = \$ \ll 144 - 6=138 \gg 138$. — 4-step business profit problem — structurally representative multi-step
D23	Define a variable ** Let x represent the number of Chevys; How many Fords does Jim have? ** Fords: $3+2x$; Write an equation ** Total: $x+3+2x+12+8x=301$; Combine like terms ** $11x+15=301$ — Socratic config breaks algebraic solution into explicit sub-question steps — useful chain-of-thought scaffold
D25	Starting with \$100, he paid \$20 to Colin, leaving him with $\$100 - \$20 = \$ \ll 100 - 20=80 \gg 80$. Twice as much as \$20 is $2 \cdot \$20 = \$ \ll 2 \cdot 20=40 \gg 40$. Thus, he paid \$40 to Helen, leaving him with $\$80 - \$40 = \$ \ll 80 - 40=40 \gg 40$. — Clean running-total tracking across 4 sequential operations — illustrates well-formed multi-step solution

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required test case categories include misapplied operation, arithmetic slip, wrong setup/misread, incomplete reasoning chain, plus US-curriculum subtypes (fraction/decimal confusion, US customary unit conversion, place value, Common Core phrasing misread, order of operations), differentiated by grade band 3–6 (per elicitation Q1/A1 and region YAML).

IO-2: Yes. GSM8K has no error-type subtaxonomy [[Q31](#), [Q60](#)]; no grade-level differentiation across 3–6; no unit-conversion problem category found in the dataset sample ([DATASET-D3](#), [DATASET-D17](#) use customary units but require no conversion). All deployment-relevant diagnostic categories are omitted.

IO-3: Yes. Sampled problems include content above grades 3–6: algebraic equation solving ([DATASET-D8](#), [DATASET-D10](#)), simple interest [[DATASET-D9](#)], real-estate pricing [[DATASET-D28](#)]. These exercise reasoning irrelevant to the elementary diagnostic context.

IO-4: Content validity is harmed in two directions: construct underrepresentation (no error-type taxonomy, no grade-level metadata, no unit-conversion category) and construct-irrelevant variance (above-grade algebra/finance problems). The MR-GSM8K result that top GSM8K solvers collapse to MR-Scores <0.02 on error localization [[WEB-7](#), [WEB-25](#)] empirically confirms the ontology mismatch.

ID	Evidence
Q1	"we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems." (p.1)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q22	"Similar to LogiQA, which requires a mix of reading comprehension and logical reasoning, GSM8K's main difficulty lies in both properly interpreting a question and reasoning through the steps to solve it." (p.4)
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
D3	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Uses US customary units (feet, cubic feet) in a volume problem
D8	Let x represent the number of Chevys; Fords: $3+2x$; Buicks: $4(3+2x)=12+8x$; Total: $x+3+2x+12+8x=301$; $11x+15=301$ — Uses algebraic variables and simultaneous-style equation — above elementary grade band (grades 3–6)
D9	If she earns an interest of 10% in the first year, her savings account increases by $10/100 * \$600 = \$\ll 10/100*600=60 \gg 60$. In the second year, she earns the same amount of interest, which $\$60 + \$60 = \$\ll 60+60=120 \gg 120$ — Simple interest calculation — above typical grades 3–6 curriculum; note solution applies simple (not compound) interest correctly
D10	Let N be the original price each friend was going to pay. $10N=6(N+8)$; $10N=6N+48$; $4N=48$; $N=\ll 12=12 \gg 12$ — Algebraic equation setup with variable N — outside grades 3–6 scope
D17	In a northwestern town, it rained 4 inches per day during the first 15 days of November. For the remainder of the month, the average daily rainfall was twice the amount observed during the first 15 days. — "Inches" as a US customary unit; geographic reference to a "northwestern town" — US-situated
D28	The price is \$98 per sq ft and it's 3,400 sq ft big so the property costs $98*3400 = \$\ll 98*3400=333200.00 \gg 333,200.00$ — Real estate pricing problem with \$333,200 result — entirely adult financial domain
WEB-5	ASDiv provides problem-type taxonomy and grade-level metadata but no student-error taxonomy (https://aclanthology.org/2020.acl-main.92.pdf)
WEB-7	MR-GSM8K: top LLMs >80% on GSM8K collapse to MR-Scores <0.02 on error localization (https://github.com/JIA-Lab-research/MR-GSM8K)
WEB-25	EmergentMind summary corroborating GSM8K→meta-reasoning collapse (https://www.emergentmind.com/topics/gsm8k)

Information Gaps

- No published quantitative breakdown of GSM8K problem types by CCSS-M grade level or by unit-conversion presence.
- Frequency of Common Core phrasings ('how many more' vs. 'how many total') across the full 8.5K dataset is unquantified.

Requires Expert Verification

- Manual audit of the GSM8K problem set by a US elementary math curriculum specialist to estimate the fraction of problems in/out of grades 3–6 CCSS-M scope.

Remediation

Gap 1: No error-type taxonomy, no grade-level metadata, no unit-conversion category; substantial fraction of problems above grades 3–6 scope.

Recommendation: Build a deployment-internal evaluation set that labels each problem by CCSS-M grade band (3, 4, 5, 6), by required error-trigger category (misapplied operation, arithmetic slip, wrong setup, incomplete chain, plus

US-curriculum subtypes), and filter out algebra/finance problems above the elementary band. Use ASDiv [WEB-5] for problem-type metadata and supplement with curriculum-specialist tagging.

ID	Evidence
WEB-5	ASDiv provides problem-type taxonomy and grade-level metadata but no student-error taxonomy (https://aclanthology.org/2020.acl-main.92.pdf)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment's HIGH-priority concern is that inputs are student-authored work with carry marks, area models, partial-quotients division, number bonds, remainder shorthand, and informal phrasing. GSM8K's input content is professionally authored or GPT-3-seeded clean prose [Q9, Q103, Q104, Q110, Q111], with linguistic diversity but stylistic only, not student-register. Every sampled datapoint shows fluent, grammatical English with machine-generated calculator annotations and no student-work features (DATASET-D2, DATASET-D19). Cultural fit is partial: USD currency and US consumer references are prevalent (DATASET-D5, DATASET-D20), and US customary units appear incidentally (DATASET-D3, DATASET-D17). However, adult financial/professional contexts dominate a noticeable share of problems (DATASET-D15, DATASET-D27, DATASET-D28) and one metric-unit example appears [DATASET-D24]. The 2026 DrawEduMath follow-up empirically confirms that models 'misdiagnose errors' on student work containing errors [WEB-11], directly evidencing the IC validity concern.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q110	"To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model." (p.15)
Q111	"Contractors were allowed to use those seed questions directly, to use them as inspiration and make modifications, or to come up with their own questions entirely." (p.15)
D2	On Tuesday Buddy has $30/2 = \ll 30/2=15 \gg 15$ baseball cards. On Wednesday Buddy has $15+12 = \ll 15+12=27 \gg 27$ baseball cards. — Well-formed natural language solution with calculator annotation format; no student notation
D3	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Uses US customary units (feet, cubic feet) in a volume problem
D5	They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. How much did the unicorn piñata and the treats cost altogether? — US-branded consumer products; implicitly assumes familiarity with US candy brands
D15	First find the increase in rent by multiplying last year's rent by 30%: $\$1000 * .3 = \$\ll 1000*.3=300 \gg 300$. Then find the food cost increase by multiplying last year's costs by 50%: $\$200 * .5 = \$\ll 200*.5=100 \gg 100$. — Multi-operation percentage problem with adult financial context (rent, insurance) — above grades 3–6 child context
D17	In a northwestern town, it rained 4 inches per day during the first 15 days of November. For the remainder of the month, the average daily rainfall was twice the amount observed during the first 15 days. — "Inches" as a US customary unit; geographic reference to a "northwestern town" — US-situated
D19	First find the total number of birds eaten per day: $30 \text{ snakes} * 3 \text{ birds/snake} = \ll 30*3=90 \gg 90$ snakes. Then multiply the number of snakes by the number of beetles per snake to find the total number of beetles eaten per day: $90 \text{ snakes} * 12 \text{ beetles/snake} = \ll 90*12=1080 \gg 1080$ beetles — Solution contains a labeling error: "90 snakes" should be "90 birds"; illustrates that correct final answers can have flawed intermediate narration
D20	The cost of the ice cream is $10 \times \$4 = \$\ll 10*4=40 \gg 40$. The cost of the frozen yoghurt is $4 \times \$1 = \$\ll 4*1=4 \gg 4$. Caleb spent $\$40 - \$4 = \$36$ more on ice cream than on frozen yogurt. — "How much more" subtraction problem — matches Common Core phrasing; solved correctly

D24	Second person = $27 - 7 = 20$ kg; $103 - 27 - 20 = 56$ kg; $56/2 = 28$ kg; The last two people each lost 28 kilograms of weight. — Uses metric units (kilograms) — not US customary; one of the few non-US-unit examples in the sample
D27	He sold his car for $20000 \times .8 = 16000$. He bought the new car for $30,000 \times .9 = 27,000$. That means he was out of pocket $27,000 - 16,000 = 11,000$ — Adult financial scenario (car purchase, percentages of \$20,000–\$30,000) — well outside elementary student context
D28	The price is \$98 per sq ft and it's 3,400 sq ft big so the property costs $98 \times 3400 = 333,200$ — Real estate pricing problem with \$333,200 result — entirely adult financial domain
WEB-11	2026 DrawEduMath follow-up: VLMs misdiagnose errors specifically on student work containing errors (https://arxiv.org/pdf/2603.00925)

Strengths

- USD currency and US consumer/school contexts dominate, partially matching the deployment's US frame (DATASET-D5, DATASET-D20).
- Some incidental coverage of US customary units (feet, pounds, inches) in problem givens (DATASET-D3, DATASET-D17).
- 'How many more' phrasing — a named CCSS-M-aligned error trigger — appears operationalized correctly [DATASET-D20].
- GSM8K is designed for high linguistic diversity within elementary content [Q9], giving some surface-form variation.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
D3	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Uses US customary units (feet, cubic feet) in a volume problem
D5	They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. How much did the unicorn piñata and the treats cost altogether? — US-branded consumer products; implicitly assumes familiarity with US candy brands
D17	In a northwestern town, it rained 4 inches per day during the first 15 days of November. For the remainder of the month, the average daily rainfall was twice the amount observed during the first 15 days. — "Inches" as a US customary unit; geographic reference to a "northwestern town" — US-situated
D20	The cost of the ice cream is $10 \times \$4 = \40 . The cost of the frozen yoghurt is $4 \times \$1 = \4 . Caleb spent $\$40 - \$4 = \$36$ more on ice cream than on frozen yogurt. — "How much more" subtraction problem — matches Common Core phrasing; solved correctly

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires parsing US-customary-unit references, Common Core phrasing conventions, and US child-context problem framings. Region YAML and elicitation Q3/A3 specify these as input features.

IC-2: Partially. USD and US contexts align (DATASET-D5, DATASET-D20), but adult-finance and real-estate contexts (DATASET-D15, DATASET-D27, DATASET-D28) and a metric-unit instance [DATASET-D24] reduce alignment with the

elementary student frame.

IC-3: Yes — algebraic and adult financial contexts ([DATASET-D8](#), [DATASET-D10](#), [DATASET-D27](#), [DATASET-D28](#)) require knowledge or reasoning patterns outside grades 3–6 students' frame.

IC-4: Regional annotation by US elementary teachers would be needed to verify (a) the fraction of GSM8K problems within grades 3–6 CCSS-M scope and (b) the frequency of US-curriculum-specific phrasing and unit-conversion items.

INSUFFICIENT DOCUMENTATION on this from the paper alone.

IC-5: Content validity is harmed primarily by the complete absence of student-register notation (carry marks, area models, partial quotients, number bonds, 'r' remainder shorthand, informal phrasing). DrawEduMath empirical findings ([WEB-11](#)) indicate that models fail specifically where the deployment most needs them to succeed.

ID	Evidence
Q3	"At test time, we generate many candidate solutions and select the one ranked highest by the verifier." (p.1)
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q21	"Similar to CommonsenseQA, GSM8K includes questions that require basic background knowledge, like the number of days in a week." (p.4)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q110	"To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model." (p.15)
D5	They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. How much did the unicorn piñata and the treats cost altogether? — US-branded consumer products; implicitly assumes familiarity with US candy brands
D8	Let x represent the number of Chevys; Fords: $3+2x$; Buicks: $4(3+2x)=12+8x$; Total: $x+3+2x+12+8x=301$; $11x+15=301$ — Uses algebraic variables and simultaneous-style equation — above elementary grade band (grades 3–6)
D10	Let N be the original price each friend was going to pay. $10N=6(N+8)$; $10N=6N+48$; $4N=48$; $N=12$ — Algebraic equation setup with variable N — outside grades 3–6 scope
D15	First find the increase in rent by multiplying last year's rent by 30%: $\$1000 * .3 = \300 . Then find the food cost increase by multiplying last year's costs by 50%: $\$200 * .5 = \100 . — Multi-operation percentage problem with adult financial context (rent, insurance) — above grades 3–6 child context
D20	The cost of the ice cream is $10 \times \$4 = \40 . The cost of the frozen yoghurt is $4 \times \$1 = \4 . Caleb spent $\$40 - \$4 = \$36$ more on ice cream than on frozen yogurt. — "How much more" subtraction problem — matches Common Core phrasing; solved correctly
D24	Second person = $27 - 7 = 20$ kg; $103 - 27 - 20 = 56$ kg; $56/2 = 28$ kg; The last two people each lost 28 kilograms of weight. — Uses metric units (kilograms) — not US customary; one of the few non-US-unit examples in the sample
D27	He sold his car for $20000 * .8 = \$16000$. He bought the new car for $30,000 * .9 = \$27000$. That means he was out of pocket $27,000 - 16,000 = \$11,000$. — Adult financial scenario (car purchase, percentages of \$20,000–\$30,000) — well outside elementary student context
D28	The price is \$98 per sq ft and it's 3,400 sq ft big so the property costs $98 * 3400 = \$333,200$. — Real estate pricing problem with \$333,200 result — entirely adult financial domain
WEB-11	2026 DrawEduMath follow-up: VLMs misdiagnose errors specifically on student work containing errors (https://arxiv.org/pdf/2603.00925)

Information Gaps

- Frequency distribution of US customary unit references and unit-conversion problems across the full dataset.
- Fraction of GSM8K problems whose contexts (school, household, child activities) match grades 3–6 student frame versus adult-domain contexts.

Requires Expert Verification

- US elementary educator review to flag which GSM8K problems contain context, phrasing, or numeric scale outside the grade 3–6 student frame.

Remediation

Gap 1: GSM8K contains no student-register notation (carry marks, area models, partial quotients, number bonds, 'r' shorthand, informal phrasing); adult-domain contexts overrepresented.

Recommendation: Collect or license a corpus of grades 3–6 student-authored arithmetic work — DrawEduMath text-derivative annotations [WEB-9, WEB-10] for vision-derived work, or Daheim et al. (2024) [WEB-13] for text-format student reasoning chains — and use this as the primary IC evaluation set. Treat GSM8K as out-of-distribution input.

ID	Evidence
WEB-9	DrawEduMath teacher-annotated student work (CCSS-aligned) (https://arxiv.org/abs/2501.14877)
WEB-10	DrawEduMath benchmark site (https://drawedumath.org/)
WEB-13	Daheim et al. (2024) ~1K student-reasoning chains with first-error annotation (https://arxiv.org/pdf/2407.09136)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form priority is MODERATE per elicitation. Both benchmark and deployment are English text [Q9, Q32, Q33], so there is no gross modality mismatch — a meaningful partial alignment. However, GSM8K inputs are well-formed prose sentences with machine-injected `<<expr=result>>` calculator annotations [Q40, Q41, Q119, DATASET-D2], whereas student-submitted work includes vertical algorithms with carry marks, area models, partial-quotients layouts, number bonds, 'r' remainder shorthand, and informal phrasing — none of which appear anywhere in the sampled dataset. The form mismatch is therefore narrower than for IC (it's a representational rather than ontological gap) but still material: a parser tuned on GSM8K-style prose has no exposure to the quasi-structured notation it will encounter.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q33	"Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans." (p.5)
Q40	"To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set." (p.6)
Q41	"At test time, a calculator will override sampling when the model chooses to use these annotations." (p.6)
Q119	"During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens." (p.17)
D2	On Tuesday Buddy has $30/2 = \text{<<30/2=15>>}$ 15 baseball cards. On Wednesday Buddy has $15+12 = \text{<<15+12=27>>}$ 27 baseball cards. — Well-formed natural language solution with calculator annotation format; no student notation

Strengths

- Shared text+English modality eliminates gross signal mismatch (no audio, no image, no multilingual issues) [Q9, Q32, Q33].
- Multi-step natural-language solutions are demanded — the paper shows direct-answer finetuning drops to 5.2% from 20.6% [Q57], so intermediate-step text is intrinsic to the form.
- Socratic config's labeled sub-question format [DATASET-D23] provides a structural scaffold partially analogous to step-indexed work.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q33	"Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans." (p.5)
Q57	"If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%." (p.7)

D23	Define a variable ** Let x represent the number of Chevys; How many Fords does Jim have? ** Fords:3+2x; Write an equation ** Total:x+3+2x+12+8x=301; Combine like terms ** 11x+15=301 — Socratic config breaks algebraic solution into explicit sub-question steps — useful chain-of-thought scaffold
-----	---

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both are English text. However, deployment text includes quasi-structured notation (vertical algorithms, area models, partial products, arrow flow) absent from GSM8K's clean prose [DATASET-D2]. Calculator annotation tokens ``<<...>>`` in GSM8K solutions [Q40, Q119] have no counterpart in student work.

IF-2: US K-12 device landscape (Chromebooks, iPads, browser-capable) supports text-based deployment per [WEB-21, WEB-22, WEB-23, WEB-24]. No infrastructure barrier prevents text input.

IF-3: Domain-specific differences: student notation includes carry marks, crossing-out, area-model layout, partial-quotients division, number bonds, 'r' remainder shorthand, and informal phrasing (per region YAML); none appear in GSM8K's input form.

IF-4: External validity is moderately harmed: the benchmark's input form is a proper subset of the form distribution the deployment encounters, so good GSM8K performance does not establish robustness to notational heterogeneity.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q40	"To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set." (p.6)
Q57	"If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%." (p.7)
Q119	"During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens." (p.17)
D2	On Tuesday Buddy has $30/2 = \text{<<30/2=15>>}$ 15 baseball cards. On Wednesday Buddy has $15+12 = \text{<<15+12=27>>}$ 27 baseball cards. — Well-formed natural language solution with calculator annotation format; no student notation
WEB-21	Chromebooks in 93% of US school districts (2025) (https://www.aboutchromebooks.com/chromebooks-in-schools-statistics/)
WEB-22	iPad significant in lower elementary; Chromebooks from grade 2 upward (https://www.edtechupdate.com/chromebook/ipad/survey/)
WEB-23	88% of public schools have 1-to-1 computing programs (2024–25) (https://ies.ed.gov/learn/press-release/more-half-public-)
WEB-24	https://www.monstermath.app/blog/how-many-us-schools-use-online-learning-2026

Information Gaps

- Whether deployment preprocessing strips calculator annotations or otherwise normalizes representational differences is INSUFFICIENT DOCUMENTATION.

Requires Expert Verification

- Deployment-team verification of whether OCR/handwriting preprocessing is in scope and how unconventional notation is canonicalized before reaching the LLM.

Remediation

Gap 1: Quasi-structured student notation (vertical algorithms, area models, partial quotients) has no analog in GSM8K's clean-prose form.

Recommendation: Define an explicit preprocessing/parsing layer for student notation forms in scope (per elicitation Q3/A3) and build form-robustness tests using perturbed or synthesized student-style inputs. Document which notation forms the LLM is expected to parse natively versus which require canonicalization.

ID	Evidence
Q3	"At test time, we generate many candidate solutions and select the one ranked highest by the verifier." (p.1)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HIGH priority and the most severe mismatch. GSM8K's output ontology is strictly binary at the solution level — correct vs. incorrect, judged solely on whether the final numerical answer matches the gold answer [Q31, Q60]. Even the token-level verifier outputs a scalar value function, not error-type labels [Q74, Q78]. The paper explicitly notes that solutions reaching the correct answer through flawed reasoning are still labeled correct [Q61], confirming that the ontology cannot distinguish reasoning quality from final-answer correctness. Dataset inspection confirms records contain only {question, answer} with no error-location, error-type, or grade-level acceptability metadata (DATASET-D1, DATASET-D2, DATASET-D16). The deployment requires structured output identifying earliest error location, independent downstream errors, error-type classification across five+ categories, and teacher-register explanation. The MR-GSM8K empirical result — top GSM8K solvers collapse to MR-Scores <0.02 on error localization [WEB-7, WEB-25] — quantifies the gap as not merely structural but severe.

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q61	"In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives." (p.7)
Q74	"This can be viewed as a token-level value function." (p.9)
Q78	"We hypothesize that the full value function provides a useful auxiliary signal that encourages the model to judge the reasoning throughout solutions, rather than merely memorizing the correct final answer." (p.9)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Clean, multi-step prose word problem requiring 4 sequential arithmetic operations
D2	On Tuesday Buddy has $30/2 = \ll 30/2=15 \gg 15$ baseball cards. On Wednesday Buddy has $15+12 = \ll 15+12=27 \gg 27$ baseball cards. — Well-formed natural language solution with calculator annotation format; no student notation
D16	If the bakery can make 12 pies, this means there would be $12 * 3 = \ll 12*3=36 \gg 36$ pie pieces. For all the pieces the bakery would make $36 * 4 = \$\ll 36*4=144 \gg 144$. The cost of making 12 pies is $12 * 0.5 = \$\ll 12*0.5=6 \gg 6$. That means the bakery would make $144 - 6 = \$\ll 144-6=138 \gg 138$. — 4-step business profit problem — structurally representative multi-step
WEB-7	MR-GSM8K: top LLMs >80% on GSM8K collapse to MR-Scores <0.02 on error localization (https://github.com/JIA-Lab-research/MR-GSM8K)
WEB-25	EmergentMind summary corroborating GSM8K→meta-reasoning collapse (https://www.emergentmind.com/topics/gsm8k)

Strengths

- Token-level verifier produces per-token scalar value predictions [Q74, Q147, Q148] — the closest GSM8K comes to step-level signal, useful as an interpretability artifact even though it isn't evaluated.
- Final-answer correctness, while too narrow, is a necessary sanity-check signal that the deployment's symbolic checker also uses (per region YAML).

ID	Evidence
----	----------

Q74	"This can be viewed as a token-level value function." (p.9)
Q147	"One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples." (p.21)
Q148	"Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set." (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: GSM8K's label categories (correct, incorrect) are deployment-relevant only as a baseline numerical sanity check; they do not cover error location, error type, or grade-level acceptability.

OO-2: Missing categories: earliest error step, independent downstream errors, five error types (misapplied operation, arithmetic slip, wrong setup, incomplete chain), US-curriculum subtypes (fraction/decimal confusion, unit conversion, place value, Common Core phrasing misread, order of operations), and grade-level acceptability differentiation. All absent [[Q31](#), [Q60](#), [Q61](#)].

OO-3: The paper acknowledges that solutions with flawed reasoning but correct final answers are labeled correct [[Q61](#)] — this encodes a final-answer-only value that is misaligned with the deployment's pedagogical-judgment ground truth.

OO-4: Stakeholder-driven taxonomy redesign is required. ProcessBench [[WEB-14](#), [WEB-15](#)] and MR-GSM8K [[WEB-7](#)] partially address earliest-error-location. BEA 2025 Shared Task [[WEB-16](#)] (Mistake Identification, Mistake Location, Providing Guidance, Actionability) is the most aligned framework.

OO-5: Structural validity is severely compromised: the construct's structure (multi-dimensional diagnostic judgment) is misrepresented as a single binary scalar. Content validity is compromised by complete category omission. External validity is compromised — confirmed by the >80%-to-<0.02 collapse [[WEB-7](#), [WEB-25](#)].

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q61	"In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives." (p.7)
WEB-7	MR-GSM8K: top LLMs >80% on GSM8K collapse to MR-Scores <0.02 on error localization (https://github.com/JIA-Lab-research/MR-GSM8K)
WEB-14	ProcessBench: 3,400 cases with human-annotated earliest erroneous step, including 400 GSM8K-level cases (https://arxiv.org/pdf/2412.06559)
WEB-15	https://huggingface.co/datasets/Qwen/ProcessBench
WEB-16	BEA 2025 Shared Task: four-dimensional tutor-response evaluation framework (https://arxiv.org/pdf/2505.18549)
WEB-25	EmergentMind summary corroborating GSM8K→meta-reasoning collapse (https://www.emergentmind.com/topics/gsm8k)

Requires Expert Verification

- Whether ProcessBench-style step-level annotation can be adapted to student-authored (not LLM-generated) work for the deployment's grade band.

Remediation

Gap 1: Binary final-answer correctness cannot evaluate earliest-error localization, error-type classification, or grade-level acceptability.

Recommendation: Adopt ProcessBench [WEB-14] for earliest-error-step evaluation as a supplementary benchmark; adopt the BEA 2025 four-dimensional framework [WEB-16] (Mistake Identification, Mistake Location, Providing Guidance, Actionability); and build a deployment-specific error-type classification eval against the five+ categories with grade-differentiated rubrics.

ID	Evidence
WEB-14	ProcessBench: 3,400 cases with human-annotated earliest erroneous step, including 400 GSM8K-level cases (https://arxiv.org/pdf/2412.06559)
WEB-16	BEA 2025 Shared Task: four-dimensional tutor-response evaluation framework (https://arxiv.org/pdf/2505.18549)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HIGH priority and severely misaligned. GSM8K ground-truth labels are produced by Upwork/Surge AI freelance contractors who re-solved problems and compared final numerical answers [Q103, Q104, Q105, Q106], with an estimated 1.7% breaking-error/ambiguity rate and acknowledged subtle errors [Q107, Q108, Q109]. No domain educators, curriculum specialists, or classroom teachers participated. Annotator demographics are not documented. Calculator annotations were generated by hard-coded logic and a finetuned language model, not human contractors [Q117, Q118]. Dataset inspection reveals concrete annotation defects (DATASET-D4: $6 \times 2 = \ll 6 + 6 = 12 \gg$ inconsistency) and at least one likely problem-setup error that survived review (DATASET-D22: 'every sixth customer' but $\div 5$). The deployment's ground truth comes from former US elementary/middle school teachers and curriculum specialists applying grade-differentiated pedagogical standards (elicitation Q4/A4) — a fundamentally different annotator population and judgment type.

ID	Evidence
Q4	"We demonstrate that verification significantly improves performance on GSM8K, and we provide strong empirical evidence that verification scales more effectively with increased data than a finetuning baseline." (p.1)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
Q117	"The calculator annotations were not provided by human contractors: they were generated by a combination of hard-coded logic and a finetuned language model." (p.17)
Q118	"The logic for auto-generating calculator annotations is imperfect. It is highly unlikely to generate any incorrect annotations, but it is not uncommon for it to ignore some lines that could be annotated." (p.17)
D4	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9 to reach a weight of $6 \times 2 = \ll 6 + 6 = 12 \gg$ 12 pounds. — Answer annotation contains an error in the expression: $\ll 6 + 6 = 12 \gg$ should be $\ll 6 \times 2 = 12 \gg$; confirms the paper's acknowledgment of annotation imperfections
D22	He sold 50 cones because $100 / 2 = \ll 100 / 2 = 50 \gg 50$. He gave away 10 cones because $50 / 5 = \ll 50 / 5 = 10 \gg 10$ — Note: "every sixth customer" → solution divides by 5, not 6. Possible arithmetic/logic error in problem or solution

Strengths

- Two-stage re-solve quality control with disagreement resolution [Q105, Q106] gives a baseline final-answer correctness signal.
- Pairwise similarity checks for template re-use [Q112, Q113] reduce one form of construct-irrelevant variance.

- Estimated breaking-error rate (~1.7%) is reported, providing at least some transparency [Q107, Q108].

ID	Evidence
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q112	"We instructed contractors to be as descriptive as possible in their solutions, and to not re-use problem settings or templates between different questions." (p.15)
Q113	"To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No. Ground-truth labels reflect contractor agreement on final numerical answers [Q31, Q60], not US elementary teachers' grade-differentiated pedagogical judgment (elicitation Q4/A4).

OC-2: High potential for disagreement. The deployment requires labels like 'acceptable as an improper fraction at grade 5 but flagged at grade 3' (region YAML); GSM8K labels carry no such grade-sensitive information. JEDM corpus [WEB-17] shows teacher-validated annotation at scale is feasible but absent from GSM8K.

OC-3: Annotator demographics are not documented beyond 'Upwork freelance contractors' and 'Surge AI' [Q102, Q103, Q104]. No data statement or datasheet was located. INSUFFICIENT DOCUMENTATION on annotator demographics, training, or domain background.

OC-4: Re-annotation by representative regional annotators (former US elementary teachers + curriculum specialists, per elicitation Q4/A4) is required for any deployment use. DrawEduMath's teacher QA layer [WEB-9, WEB-10] is a precedent.

OC-5: Aggregation method (final-answer agreement; repair or discard on disagreement [Q106]) collapses pedagogical disagreement to binary signal, which would erase minority/edge-case judgments. No reporting of inter-annotator agreement beyond the 1.7% breaking-error estimate [Q107].

OC-6: Convergent validity harmed: labels do not correlate with what classroom teachers would judge for the same student work. External validity harmed: contractor agreement on final answer does not generalize to pedagogical judgment.

ID	Evidence
Q4	"We demonstrate that verification significantly improves performance on GSM8K, and we provide strong empirical evidence that verification scales more effectively with increased data than a finetuning baseline." (p.1)
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)

Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q61	"In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives." (p.7)
Q102	"We thank Dan Hendrycks, Leo Gao, Alec Radford, and Giambattista Parascandolo for their valuable feedback on this paper; Harri Edwards, Yura Burda, Michael Wu, and Nick Ryder for many insightful conversations; Michael Petrov, Alethea Power, and Jacob Jackson for their technical assistance; the OpenAI Supercomputing team for the infrastructure that made these experiments possible; and the team at Surge AI for performing the GSM8K data collection." (p.12)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
WEB-9	DrawEduMath teacher-annotated student work (CCSS-aligned) (https://arxiv.org/abs/2501.14877)
WEB-10	DrawEduMath benchmark site (https://drawedumath.org/)
WEB-17	JEDM 8,307 problems / 193,187 student responses / 1,296 teachers with 5-point scale + free-text feedback — precedent for teacher-validated ground truth at scale (https://jedm.educationaldatamining.org/index.php/JEDM/article/download/858/258)

Information Gaps

- No documented annotator demographics, training, or curriculum-domain background for GSM8K contractors.
- No inter-annotator agreement statistic beyond the 1.7% breaking-error estimate.

Requires Expert Verification

- Internal deployment team's IAA figures (region YAML notes these are [NEEDS VERIFICATION]).
- US elementary teacher review of GSM8K labels for pedagogical-acceptability conflicts at grade 3, 4, 5, and 6 norms.

Remediation

Gap 1: Ground truth is freelance-contractor final-answer agreement, with no domain educators, no grade-level differentiation, no documented annotator demographics, and a 1.7% breaking-error rate plus acknowledged subtle errors.

Recommendation: Construct an in-house teacher-annotated evaluation set with the deployment's stated annotator pool (former US elementary/middle school teachers + curriculum specialists). Use the JEDM corpus methodology [WEB-17] (5-point scale + free-text feedback) and DrawEduMath teacher-QA layer [WEB-10] as design precedents. Report IAA by grade level.

ID	Evidence
WEB-10	DrawEduMath benchmark site (https://drawedumath.org/)
WEB-17	JEDM 8,307 problems / 193,187 student responses / 1,296 teachers with 5-point scale + free-text feedback — precedent for teacher-validated ground truth at scale (https://jedm.educationaldatamining.org/index.php/JEDM/article/download/858/258)

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GSM8K | Context: US Elementary/Middle School Arithmetic Tutoring — Grade 3–6 Diagnostic Module

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MODERATE priority but still substantially misaligned. GSM8K's scoring output is a binary correct/incorrect judgment derived from final-answer comparison, or in verifier mode a ranked list of completions [Q3, Q29, Q46, Q68]. The token-level visualizations are interpretability aids, not scored elements [Q147, Q150, Q151]. The deployment requires free-form structured diagnostic text in classroom-teacher register identifying earliest error, downstream errors, error type, and a student-facing explanation (elicitation Q2/A2, region YAML). MathTutorBench [WEB-18] and the 2024 AI Tutor Evaluation Taxonomy paper [WEB-19] confirm that no standard NLG metric captures pedagogical-explanation quality and that generic metrics (BLEU/BERTScore) can be gamed — meaning GSM8K's evaluation harness cannot be repurposed as a proxy. Both deployment and benchmark share text modality, which is the basis for the score not being 1.

ID	Evidence
Q2	"To increase performance, we propose training verifiers to judge the correctness of model completions." (p.1)
Q3	"At test time, we generate many candidate solutions and select the one ranked highest by the verifier." (p.1)
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q46	"Test performance is determined by a single low temperature ($T = 0$) sample for each test problem." (p.6)
Q68	"At test time, we sample 100 completions to each test problem, rank them with the verifier, and then return the one with the highest verifier score." (p.8)
Q147	"One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples." (p.21)
Q150	"The second column of the table summarizes the verifier's prediction, and the third column indicates whether the generated model completion was actually correct or incorrect." (p.22)
Q151	"Any disagreement between the second and third columns indicates that the verifier made an error." (p.22)
WEB-18	MathTutorBench: BLEU/BERTScore do not capture pedagogical value; no unified pedagogical evaluation metric exists (https://arxiv.org/html/2502.18940v1)
WEB-19	AI Tutor Evaluation Taxonomy: NLG metrics gamed by trivial responses (https://arxiv.org/html/2412.09416v1)

Strengths

- Shared text modality means deployment outputs can at least be generated in the same channel that GSM8K solutions occupy (no TTS or modality bridge needed) [Q32, Q33].
- Hyperparameter robustness is documented within the binary-accuracy paradigm [Q114, Q115, Q116] — useful only for the final-answer sanity-check sub-task.
- The paper produced human-interpretable natural-language solutions explicitly [Q32, Q33], so there is at least a free-form text artifact (though unscored for quality).

ID	Evidence
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)

Q33	"Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans." (p.5)
Q114	"We performed sweeps of the learning rate and batch size by an order of magnitude in both directions from the values in the table and were unable to find any significant improvements." (p.16)
Q115	"Other reasonable choices for both the verifier temperature (eg: 1.0 instead of 0.7) and objective (cross-entropy instead of mean squared error) also had negligible effect in our ablations." (p.16)
Q116	"Hyperparameters used for all experiments, unless explicitly said otherwise. Notable exceptions include Figure 8c, which uses 4x more tokens per batch and 300 completions at both training and test time. All dropout experiments in Figure 8 use 20% dropout. Figure 7a uses verifiers trained on 100 completions, but searching over more completions at test time." (p.16)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match. Both are text. But the required structured diagnostic format (earliest-error location + downstream flags + error-type label + teacher-register explanation, per elicitation Q2/A2) is not produced or scored by GSM8K [Q29, Q46, Q68].

OF-2: Not applicable — the deployment is text-only on browser-capable devices [WEB-21, WEB-22]; no TTS requirement.

OF-3: Target users are grades 3–6 students; explanations must be concrete and student-facing (elicitation Q2/A2). GSM8K's output form provides no register signal — only a number.

OF-4: External validity is harmed: a model's GSM8K score gives no evidence about explanation quality, register appropriateness, or pedagogical accuracy. BEA 2025 Shared Task [WEB-16] and MathTutorBench [WEB-18] are needed supplementary frameworks.

ID	Evidence
Q2	"To increase performance, we propose training verifiers to judge the correctness of model completions." (p.1)
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q46	"Test performance is determined by a single low temperature ($T = 0$) sample for each test problem." (p.6)
Q68	"At test time, we sample 100 completions to each test problem, rank them with the verifier, and then return the one with the highest verifier score." (p.8)
WEB-16	BEA 2025 Shared Task: four-dimensional tutor-response evaluation framework (https://arxiv.org/pdf/2505.18549)
WEB-18	MathTutorBench: BLEU/BERTScore do not capture pedagogical value; no unified pedagogical evaluation metric exists (https://arxiv.org/html/2502.18940v1)
WEB-21	Chromebooks in 93% of US school districts (2025) (https://www.aboutchromebooks.com/chromebooks-in-schools-statistics/)
WEB-22	iPad significant in lower elementary; Chromebooks from grade 2 upward (https://www.edtechupdate.com/chromebook/ipad/survey/)

Information Gaps

- No rubric or metric within GSM8K for evaluating teacher-register explanation quality.

Requires Expert Verification

- Whether the deployment team plans to adopt BEA 2025 / TutorBench-style rubric scoring or develop a proprietary teacher-register rubric.

Remediation

Gap 1: No evaluation mechanism for free-form teacher-register diagnostic explanation; generic NLG metrics are inadequate.

Recommendation: Use rubric-based human evaluation along the BEA 2025 dimensions [WEB-16] for register and pedagogical accuracy. Supplement with LLM-as-judge rubric scoring (TutorBench F1=0.82 vs human F1=0.91 [WEB-20]) for scale. Avoid BLEU/BERTScore as primary metrics per [WEB-18, WEB-19].

ID	Evidence
WEB-16	BEA 2025 Shared Task: four-dimensional tutor-response evaluation framework (https://arxiv.org/pdf/2505.18549)
WEB-18	MathTutorBench: BLEU/BERTScore do not capture pedagogical value; no unified pedagogical evaluation metric exists (https://arxiv.org/html/2502.18940v1)
WEB-19	AI Tutor Evaluation Taxonomy: NLG metrics gamed by trivial responses (https://arxiv.org/html/2412.09416v1)
WEB-20	TutorBench: LLM-judge with rubric criteria achieves F1=0.82 vs human F1=0.91 (https://static.scale.com/uploads/654197dc94d34f66c0f5184e/TutorBench%20(1).pdf)